

MASTER'S COMPREHENSIVE EXMINATION

Part II

Applied (90 minutes)

March 5, 2005

Instructions: Answer both questions. Open book and open notes.

QUESTION 1

In a classic physics experiment in 1798 Count Rumford measured the induced heat resulting from drilling the barrel of a cannon. Rumford also repeatedly measured the temperature of the cannon as it cooled, and plotted the following data: $\log(\text{Temp})$ vs time. He found that the relationship was near linear.

25 measurements were taken: let Temp be the temperature in Fahrenheit, and t be the time after drilling (in minutes). $\text{mean}(\log(\text{Temp}-60)) = 4.04$, $\text{mean}(t) = 21.038$, $SD(\log(\text{Temp}-60)) = .092$, $SD(t) = 12.57$, $r = -.91$.

- Find the regression line of $\log(\text{Temp})$ on time. Comment on the meaning of the regression coefficients.
- What do you expect the temperature of the cannon to be a very long time after the experiment?
- Set up 99% confidence intervals for the slope and intercept and comment.
- Set up a 99% prediction interval for the $\log(\text{Temp})$ at 30 minutes, and at 60 minutes. Comment. Sketch the regression line, and prediction intervals for various temperatures.
- Set up an approximate prediction confidence interval for the temperature Temp at 30 minutes.